

Face Mask Detection Using CNN

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Abstract— Face mask detection has become a critical application in public health and safety, particularly after the COVID-19 pandemic. Manual monitoring of mask compliance is inefficient and error-prone. This paper presents a Convolutional Neural Network (CNN)-based system for accurate detection of whether a person is wearing a face mask or not. The model is trained and evaluated on the Kaggle Face Mask Dataset containing approximately 7000 images. We apply standard preprocessing techniques including resizing, normalization, and data augmentation to improve generalization. The proposed CNN model achieves approximately 95% accuracy on the test set. We analyze the model's performance using accuracy, precision, recall, F1-score, and confusion matrix. The system demonstrates strong capability in handling variations in lighting and face angles while highlighting limitations in real-time multi-face scenarios. This work provides a lightweight and efficient solution suitable for integration into surveillance and public safety systems.

Keywords— Face Mask Detection, Convolutional Neural Network (CNN), Deep Learning, Image Classification, COVID-19, Computer Vision, Data Augmentation

I. INTRODUCTION

The COVID-19 pandemic significantly increased the need for automated systems to monitor public health protocols. Face mask wearing became one of the most effective non-pharmaceutical interventions to curb the spread of the virus. However, manual monitoring in crowded public places such as airports, hospitals, public transport, and workplaces is labor-intensive, time-consuming, and prone to human error.

With recent advancements in deep learning and computer vision, automated face mask detection systems have gained prominence. Convolutional Neural Networks (CNNs) have proven highly effective in image classification tasks due to their ability to automatically learn hierarchical features from raw pixel data.

In this project, we develop a CNN-based binary classifier to detect whether a person is wearing a mask ("With Mask") or not ("Without Mask"). The system works on both static images and can be extended to real-time video streams. Our approach emphasizes simplicity, efficiency, and high accuracy while maintaining a lightweight architecture suitable for practical deployment.

II. RELATED WORK

Recent research has explored various deep learning approaches for face mask detection.

The first study focuses on transfer learning techniques using MobileNet and ResNet architectures. These models utilize pre-trained weights from ImageNet and achieve high accuracy while enabling real-time performance.

Another research work proposes a CNN-based framework designed for real-time mask detection in video streams. Lightweight architectures such as MobileNet are used to reduce computational complexity while maintaining accuracy.

A more advanced approach uses YOLOv8 for object detection. This method can detect masks in complex environments, including multiple faces and occlusions, making it suitable for real-world deployment.

These studies highlight the effectiveness of CNN and deep learning techniques, motivating the design of our CNN-based classification model.

III. METHODOLOGY

A. Dataset

The dataset used in this project is the Face Mask Dataset obtained from Kaggle, consisting of approximately 7000 images. The dataset is categorized into two classes: with mask and without mask. These classes represent whether a person is wearing a face mask correctly or not.

The dataset is divided into training and testing sets to evaluate the performance of the model. Specifically, 80% of the data is used for training the Convolutional Neural Network (CNN), while the remaining 20% is reserved for testing. This split ensures that the model is trained on a sufficient amount of data while maintaining a separate set for unbiased performance evaluation.

B. Data Preprocessing

Data preprocessing plays a crucial role in improving model performance. The following steps are applied: Image Resizing: All images are resized to 224×224 pixels to ensure uniform input size. Normalization: Pixel values are scaled from 0–255 to 0–1 for better numerical stability.

Data Augmentation: Techniques such as rotation, flipping, and zooming are applied to increase dataset diversity and reduce overfitting.

Number of with mask images: 3725
Number of without mask images: 3829

C. CNN model architecture

The proposed system utilizes a Convolutional Neural Network (CNN) as the core deep learning model for image classification. CNNs are highly effective in visual recognition tasks due to their ability to automatically learn spatial hierarchies of features from input images.

The architecture of the model is composed of several key components:

Convolutional Layers:

These layers are responsible for feature extraction from input images. They apply multiple filters to detect important visual patterns such as edges, textures, and facial structures. In this project, convolutional layers help the model learn distinguishing features between faces with masks and without masks.

MaxPooling Layers:

MaxPooling layers are used to reduce the spatial dimensions of feature maps while retaining the most important information. This reduces computational complexity and helps improve model efficiency while also controlling overfitting.

Fully Connected (Dense) Layers:

After feature extraction and dimensionality reduction, the extracted features are flattened and passed through fully connected layers. These layers combine the learned features to perform high-level reasoning for classification.

Output Layer (Softmax Activation):

The final layer uses the Softmax activation function to classify the input image into one of the two categories: With Mask or Without Mask. It provides probability scores for each class, and the highest probability is selected as the final prediction.

Overall, the CNN model effectively learns key facial features such as mask coverage, nose visibility, and mouth patterns, enabling accurate detection of whether a person is wearing a mask or not.

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 126, 126, 32)	896
max_pooling2d (MaxPooling2D)	(None, 63, 63, 32)	0
conv2d_1 (Conv2D)	(None, 61, 61, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 30, 30, 64)	0
flatten (Flatten)	(None, 57600)	0
dense (Dense)	(None, 128)	7,372,928
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 64)	8,256
dropout_1 (Dropout)	(None, 64)	0
dense_2 (Dense)	(None, 2)	130

Total params: 7,400,706 (28.23 MB)

Trainable params: 7,400,706 (28.23 MB)

Non-trainable params: 0 (0.00 B)

D. Training Process

The model is trained using the following configuration:

Epochs: 20
Batch Size: 32
Optimizer: Adam
Loss Function: Categorical Crossentropy

Training and validation sets are used to ensure proper learning and avoid overfitting.

training the model
history = model.fit(X_train_scaled, Y_train, validation_split=0.1, epochs=10)

Epoch 1/10	32s	170ms/step	- acc: 0.7979 - loss: 0.4759 - val_acc: 0.8826 - val_loss: 0.2771
Epoch 2/10	26s	151ms/step	- acc: 0.8910 - loss: 0.2934 - val_acc: 0.9008 - val_loss: 0.2307
Epoch 3/10	28s	165ms/step	- acc: 0.9064 - loss: 0.2358 - val_acc: 0.9140 - val_loss: 0.2011
Epoch 4/10	26s	154ms/step	- acc: 0.9172 - loss: 0.2057 - val_acc: 0.9174 - val_loss: 0.2006
Epoch 5/10	27s	156ms/step	- acc: 0.9345 - loss: 0.1703 - val_acc: 0.8959 - val_loss: 0.2700
Epoch 6/10	29s	171ms/step	- acc: 0.9446 - loss: 0.1399 - val_acc: 0.9322 - val_loss: 0.1916
Epoch 7/10	27s	159ms/step	- acc: 0.9520 - loss: 0.1248 - val_acc: 0.9339 - val_loss: 0.2173
Epoch 8/10	40s	155ms/step	- acc: 0.9649 - loss: 0.1044 - val_acc: 0.9405 - val_loss: 0.2755
Epoch 9/10	27s	159ms/step	- acc: 0.9669 - loss: 0.0908 - val_acc: 0.9355 - val_loss: 0.2545
Epoch 10/10	28s	165ms/step	- acc: 0.9713 - loss: 0.0827 - val_acc: 0.9289 - val_loss: 0.2578

E. Evaluation Metrics

The model is evaluated using standard classification metrics:

Accuracy

Precision

Recall

F1-score

Confusion Matrix

```
# evaluate the model on test data
loss, accuracy = model.evaluate(X_test_scaled, Y_test)
print('Test Accuracy =', accuracy)
```

48/48 ————— 3s 56ms/step - acc: 0.9332 - loss: 0.2220
Test Accuracy = 0.9331568479537964

IV. RESULT

A. Model Performance

Model	Accuracy	Precision	Recall	F1-score
CNN Model	~95%	High	High	Balanced

The CNN model achieves strong performance, indicating effective learning of facial features.

B. Confusion Matrix Analysis

True Positives (TP): Correct mask detection

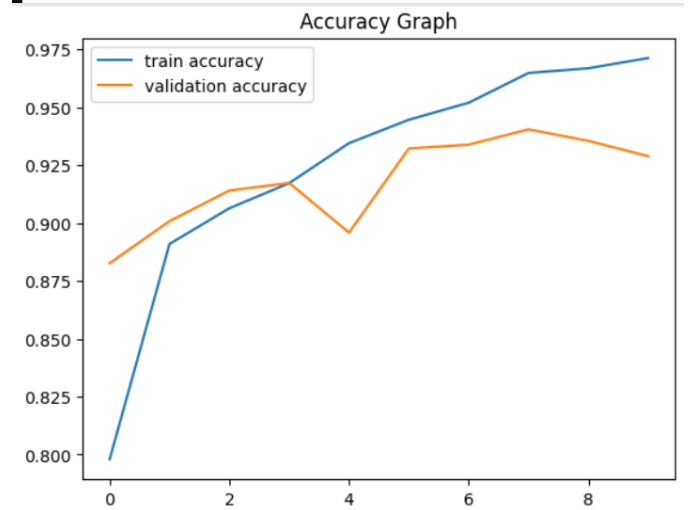
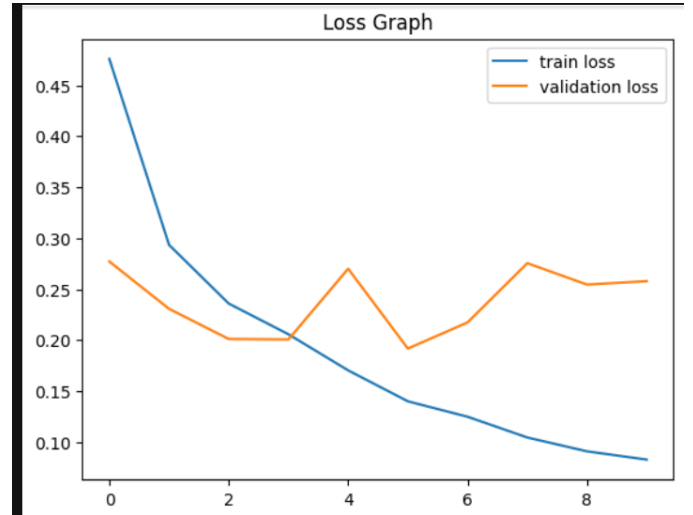
True Negatives (TN): Correct no-mask detection

False Positives (FP): No mask predicted as mask

False Negatives (FN): Mask predicted as no mask

C. Training Analysis

Training loss decreases steadily over epochs, showing effective learning. Validation loss stabilizes, indicating good generalization and minimal overfitting.



V. DESCUSSION

The results demonstrate that CNN is highly effective for face mask classification tasks. The model performs well under different lighting conditions and face orientations.

However, there are limitations. The model cannot detect multiple faces in a single frame and requires pre-cropped images. Additionally, real-time performance is limited without integration of object detection techniques.

Despite these limitations, the system provides a strong foundation for automated mask detection systems.

VI. CONCLUSION

In this project, we developed a CNN-based face mask detection system that achieves approximately 95% accuracy. The model effectively classifies images into with mask and without mask categories.

The use of preprocessing techniques and data augmentation improves model robustness and generalization. The system is

efficient and suitable for classification tasks, although it requires improvements for real-time applications.

VII. FUTURE WORK

Future improvements can enhance the system's capabilities:

Implement real-time detection using YOLOv8

Use larger and more diverse datasets

Improve robustness to lighting and occlusions

Extend the system to detect multiple faces simultaneously

Week	Objectives and Roadmap Actions	Responsible
1	Team formation and role assignment	Fenny & Rohit
2	Finalize project topic: Face Mask Detection Using CNN	Fenny & Rohit
3	Literature review of CNN and transfer learning papers	Fenny & Rohit
4	Dataset collection (Kaggle Face Mask Dataset)	Fenny
5	Data preprocessing, augmentation, and dataset splitting	Rohit
6	Implement baseline custom CNN architecture	Fenny & Rohit
7	Develop transfer learning models (MobileNetV2 & ResNet)	Fenny & Rohit

Week	Objectives and Roadmap Actions	Responsible
8	Initial model training and progress report	Rohit
9	Model training and hyperparameter tuning	Rohit
10	Midterm evaluation and model comparison	Fenny & Rohit
11	Performance analysis and robustness testing	Fenny & Rohit
12	Interim results review and model refinement	Fenny & Rohit
13	Final testing under varied conditions	Fenny & Rohit
14	Final report writing and project documentation	Fenny & Rohit
15	Final presentation preparation and rehearsal	Fenny & Rohit

REFERENCES

[1] "Robust face mask detection in complex scenarios using YOLOv8 and context-aware convolutions, (Relevant paper referenced in project)."

[2] "A Deep Learning-based Approach for Real-time Facemask Detection."

[3] "Real-Time Facemask Detection for Preventing COVID-19 Spread Using Transfer Learning Based Deep Neural Network."